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COMPARING GREEK AND CHINESE PHILOSOPHY AND SCIENCE

Introduction

Most people are aware by now that we can find science, technology, and medicine in every culture, but the outcome of decades of research on non-European traditions has not been, as we might have hoped, a narrative of world science.[1] What we find instead is a set of stories that remain largely unrelated until they end, one by one, with bodies of knowledge and practice pulverized by the impact of the West. Some bits survive, more or less, such as traditional Chinese medicine. So far as the rest of the world is concerned, they are exotic curiosities rather than parts of an understanding that hang together.

The history of science ought to be more than a collection of cultural ghettos. That is why people curious about the evolution of science as a general human phenomenon have always invested such high hopes in comparative studies.

For three hundred years historians have been forming judgments about the relations of European and other scientific traditions, and the main similarities and differences between them. But if we ask what we have learned from these three centuries of comparative studies, the answer is embarrassingly little.

Certainly no educated person today would claim that the Chinese or the Japanese had no aptitude for science, or that Islamic science depended entirely on borrowings from Greece. Every undergraduate who has taken an introductory course in the history of science knows that the centers of innovation in late-fourteenth-century mathematical astronomy were not Paris and Oxford but Seoul and Baghdad. That is not the result of comparative study, but of research by specialists on one civilization at a time.

Explicitly comparative study has called attention to a number of parallels between civilizations, and to regular contacts between their technical traditions. But has it uncovered new possibilities of human thought and practice? Has it led

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1. I use "Greece" as shorthand for a wide domain whose elites wrote Greek. It included much of the Mediterranean littoral of the Middle East. In the Hellenistic period it extended to Northern India, Egypt, and other parts of north Africa. I am grateful to Judith Zeitlin, Benjamin Schwartz, and others for an exuberant discussion when I offered this paper at the Harvard Pre-Modern China Seminar.

to substantial understanding of worldwide science? Has it helped us to a deeper comprehension of Japanese medicine, or Greek epistemology, or Indian mathematics? A certain number of facts and dates has accumulated, but the conclusions drawn in the many published comparisons seldom affect our own understanding.

Current Work

Disappointment will continue to outbalance hope as long as we insist on comparing things out of context one at a time, whether they are concepts, values, machines, or groups of people. I cannot offer a ready-made formula for a more optimistic approach, but let me describe my current explorations.

They began four years ago when I met G.E.R. Lloyd, a historian of classical philosophy and science who for some time had been as worried as I had been about the need for comparisons that made sense. He had been following recent and innovative scholarship on China, as I had done with respect to Europe. We began a series of exploratory studies that we hoped would point us in revelatory directions. After three years of exchanging what turned out to be a thick stack of drafts, and meeting in the summers to argue about them, we decided that this experiment was worth continuing. It was yielding new questions that suggested new ways to look at the traditions we studied.

We decided to begin a more or less systematic collaborative study of Chinese and Greek natural philosophy and science. At present we are rereading the primary literature, led by new questions to find new things in it. We have begun drafting and rewriting parts of what will take several years to become a finished book. Our preliminary studies, not at all restricted to science, have led us to define what seem to be several useful approaches to comparison:

1. An advantage of cooperation is that it gives us approximately equal depth in the study of the two cultures. We do not want to add to the accumulation of comparisons that put together a substantial understanding of one with superficial generalities about the other. Mastery of the languages and the primary sources of both traditions is hardly optional.

2. We do not choose what to compare by the criteria of today's science, but by what turn out to be important similarities or contrasts in Greece and China. We look for superficial similarities with different significances, or different means that lead to analogous ends. We are trying to learn how different cultural circumstances push ideas and institutions in different directions.

That leads us to draw on aspects of Greek thought that have no counterpart in China, at least in the period that interests us. An obvious example is Greek element theories, which claim that things are made up of minute ultimate parts that

usually do not look like the parts that are big enough for us to see. Element theories build on the idea that reality is hidden, and direct experience is in some ultimate sense not real. That fundamental claim, which we usually refer to as appearance vs. reality, has no counterpart in China. Exploring the circumstances in which it developed in Greece has drawn our attention to Chinese circumstances that leave no scope for such an idea.

Equally interesting, of course, are important Chinese ideas that were not found in Europe, for instance *cheng ming* 正名, which used to be translated "the rectification of names." This is actually quite a diverse group of doctrines about what should be done when names and the things that they designate do not correspond. The variations point to different social interests and political agendas. Greeks entertained vaguely similar ideas, but what we find interesting is how peripheral they were there.

Much effort has been wasted by comparativists straining to find logic in early Chinese philosophy, but no one has yet come to grips with the complementarity of Greek logic and Chinese semantics. Semantics, after all, is what the the people that historians lump together as *ming-chia* 名家 mostly discussed.[2]

More obvious comparisons are possible when concepts or assumptions turn up in both cultures. Because their settings are different, they are bound to have very different implications. Topics worth exploring range from patronage as a form of livelihood (which had very different implications in the two cultures)[3] to the idea that the state and the human body are both miniature replicas of Nature. I will summarize below our approach to this idea of macrocosm and microcosms, to make it clear that what we are doing is more than a conventional exercise in the history of ideas.

3. There is obviously some risk in picking activities to compare from dissimilar periods, where their settings and their significances may be very different. We have looked for a single period in which both cultures seem to be passing through more or less analogous transitions and in which the literatures are comparable. There is no particular reason that there should be such a period. But we believe that there is one.

It happens that between roughly 300 B.C. and A.D. 200 both China and Greece decisively moved away from a free-for-all in philosophic innovation. By the end of that time both were obsessed instead with preserving what had already been done.

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2. Logic has to do with the forms of thought or its expression, and semantics with the signification and meaning of words.

3. See Sivin 1995 and, more briefly, Chap. IV below.

This was also the period in which the sciences separated out of natural philosophy and established their own institutions and distinct literatures.

Medicine did so earlier in Greece. In the fifth century B.C., when philosophy was just developing its repertory, medicine was already a flourishing intellectual as well as practical enterprise. Still, in the Hellenistic period, from about 300 B.C. on, in medicine as well as in physics, people focussed more and more attention on the past, on what had been inherited from the ancients.

Those five hundred years give us a large and more or less commensurable literature in which to explore basic scientific issues that had not been pressing earlier. In natural science and mathematics we can confront Eudoxus, Euclid, Hipparchus and Ptolemy with the Mathematical Methods in Nine Chapters *(Chiu chang suan shu* 九章算術), the Arithmetical Canon of the Chou Gnomon *(Chou pi suan ching* 周髀算經), and the astronomical treatises of the first two Standard Histories. In medicine we can set Galen and his predecessors side by side with Chinese classics from the Mawangdui manuscripts through the Canon of Eighty-one Problems *(Huang-ti pa-shih-i nan ching* 黃帝八十一難經).

The period we are studying is especially significant in a sense that we did not realize when we began. It was an era of fundamental political change at both ends of Eurasia. The Hellenistic period marked the end of the independent Greek city state as a significant unit. The social order that had evolved thinkers such as Plato and Aristotle ceased to exist. The larger world that Alexander's empire created, the world in which Rome rose and fatally overextended itself by the year 200, was bound to redefine the scope of learning and science.

In China too, changes in society and politics paralleled a change in thought and work. The same half-millennium began with rapidly shifting constellations of power as the local potentates of the Chou dynasty wiped each other out. Over those five centuries, the first universal and centrally organized imperial state first stabilized itself, then disintegrated, and finally collapsed.

Before 300, ambitious intellectuals competed in a widespread system of princely patronage. As the Han state succeeded, they became civil servants in a single bureaucracy and eventually adherents of a single state orthodoxy. But that order faded. By A.D. 200 a position in the civil service could be as dangerous as it could be profitable, and the official philosophy was no longer being taken seriously. That leaves us with the stimulating challenge to determine how these political changes are related to the emergence of natural philosophy and science. The period is the same, and to a large extent the actors are the same.

4. My colleague and I are not comparing things or concepts, but processes, evolving activity. Like most historians of science today, we have no use for the

idea that science is one thing and its context another. We are looking at ideas, their use, and the social process that created and elaborated them, as a single phenomenon. How physicians or astronomers earned a living, how thinkers grouped themselves, in what ways they publicly disagreed, and what political significance they claimed for cosmology are just as revealing as concepts, forms of proof, and patterns of thought.

From this point of view it does not make sense to ask whether social change was the cause of scientific change, or whether philosophy changed politics. We see these as part of a single manifold of history. Our work is understanding what makes them one process. A prime example is the close linkages from the third century on between natural process, political process, and the vital processes of the human body.

There has been a great deal of writing on the development of cosmology in China, in connection with natural philosophy and the individual sciences. The yin-yang, Five Phases, and *ch'i* concepts were eventually combined to provide the framework for a new theory of resonance between Heaven and Earth on the one hand and the political realm on the other, with the body a second microcosm. A number of scholars including myself would put the final synthesis in the first century B.C., near the end of the Western Han. The document that consummated the mature synthesis of yin-yang, Five Phases, and *ch'i* now appears to be the Inner Canon of the Yellow Lord *(Huang-ti nei ching* 黃帝內經).

Thinkers from the third century on did not associate concepts of the state with cosmology and science because the empirical data forced them to do so. Rather, as I show in Chap. IV, the yin-yang and Five Phases concepts were moral and political from the start. In the long sweep of Chinese thought they remained at the same time moral, political, and physical.

Ideas of *ch'i*, yin-yang, and so on permeated all three domains because they were too politically important to pass by. What made them important is that the same people were using them to claim that the operations of the unified and centralized Ch'in-Han state were based on Nature's processes. They were arguing that resisting a state built on the unity of Nature was bound to fail as surely as resisting the four seasons was bound to fail. They were also trying to persuade an emperor not bound by constitution or law to limit his own freedom. They argued that he and his officials could keep cosmos and state aligned only by modeling government on the regularities and disciplines that governed heaven and earth.[4]

Let us look at the book that first systematically developed such parallels, Lü

4. Queen 1991, Sivin 1995. Part of the argument here summarizes the latter.

Pu-wei's 呂不韋 Springs and Autumns *(Lü shih ch'un-ch'iu* 呂氏春秋). It begins with the idea that interfering with the circulation in the human body leads to illness. It goes on first to draw out parallels in Nature, and then to point out their meaning for the state. This book was written in a chaotic time (ca. 239) when there had been no Chou king for a generation, and when no one could do more than hope that China would soon be unified:

> Human beings have 360 joints, nine body openings, and five yin and six yang systems of function. In the flesh tightness is desirable; in the blood vessels *(hsueh-mo* 血脈) free flow is desirable; in the sinews and bones solidity is desirable; in the operations of the heart and mind harmony is desirable; in the essential *ch'i* regular motion is desirable. When [these desiderata] are realized, illness has nowhere to abide, and there is nothing from which pathology can develop. When illness lasts and pathology develops, it is because the essential *ch'i* has become static.
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> Analogously, water when stagnant becomes foul; a tree when [the circulation of its *ch'i* is] stagnant becomes worm-eaten; grasses when [the circulation of their *ch'i* is] stagnant become withered (?).[5]
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> States too have their stases. When the ruler's virtue does not flow freely [i.e., if he does not appoint good officials to keep him and his subjects in touch], and the wishes of his people do not reach him, this is the stasis of a state. When the stasis of a state lasts for a long time, a hundred pathologies arise in concert, and a myriad catastrophes swarm in. The cruelty of those above and those below toward each other arises from this. The reason that the sage kings valued heroic retainers and faithful ministers is that they dared to speak directly, breaking through such stases.[6]

This union of politics, physics, and medicine was not temporary. It made its way into what you might call "the first Neo-Confucianism" (see Chap. IV). This promiscuous doctrine, much of it contrary to the teachings of Confucius, became conventional among the office-holding elite from the late second century on. Here is a sample from Abundant Dew on the Spring and Autumn Annals *(Ch'un-ch'iu fan lu* 春秋繁露, 156/130 B.C.), which designed the Han Confucian state orthodoxy:[7] "the king models himself on heaven. He models himself on its seasons and

5. *K'uai* 蕢 appears corrupt. None of the commentators offers a plausible reconstruction, nor can I.

6. *Lü shih ch'un-ch'iu, 20, Lan* 8. 5: 1373.

7. *Ch'un-ch'iu fan lu, 11:* 9b; the meaning of the last sentence remains uncertain. Cf. Queen 1991: 326–327. I accept the recommendations of the editor Su Yu 蘇輿 for reading this corrupt passage; he suggests that *shih* 施 and *chih* 治 are scribal errors for *fa* 法, and that *fa* appears for *chih.*

consummates them. He models himself on its commands and circulates them among all men. He models himself on its constant categories and uses them when initiating affairs. He models himself on its Way and thereby makes order emerge. He models himself on its will and thus commits [his realm] to benevolence."

The political focus continued into the literature of the distinct sciences that emerged in the first century B.C. It is common in the ancient books that came to be considered the founding canons of the various sciences. We can see it, for instance, in the dialogues between emperors and ministers in the medical Inner Canon and the cosmological Mathematical Canon. It recurs where we would most expect to find it, namely in technical writings for and by emperors, at least down to the medical Canon of Sagely Benefaction *(Sheng chi ching* 聖濟經) of 1118.

This passage from the Basic Questions *(Su wen* 素問), a version of the Inner Canon, is not about macrocosm and microcosm, but about the correspondence of the two microcosms, state and body. It begins as the Yellow Lord tells his minister Ch'i-po 岐伯 that he wants to hear "about the relative authority of the twelve systems of functions associated with the internal organs, about which is higher in rank and which lower." A key to medicine, in other words, is the political hierarchy of the visceral systems, seen as departments in a civil service. Ch'i-po replies:

> The cardiac system is the office of the monarch; consciousness issues from it. The pulmonary system is the office of the minister-mentors; oversight and supervision issue from it. The hepatic system is the office of the General; planning issues from it. The gall bladder system is the office of the rectifiers; decisions issue from it . . . [and so on for twelve systems of body functions associated with internal organs]. It will not do for these twelve offices to lose their coordination.
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> If the ruler is enlightened, all below him are secure. If one nourishes one's vital forces in accordance with this, one will live long and pass through life without peril. If one governs all under Heaven in accordance with this, it will be greatly prosperous.
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> If the ruler is unenlightened, the twelve offices will be endangered, so that the thoroughfares of circulation will be closed off and movement will not be free. The body *(hsing* 形) will be seriously injured. If one nourishes one's vitalities in accordance with this, the result will be calamity. One who governs all under Heaven in accordance with this will imperil his patrimony. Take care! Take care![8]

There is no need to pause over the choice of civil service posts, which relate each set of functions to the ensemble of vital processes. Ch'i-po's point is that

8. *Huang-ti nei ching su wen,* 8.1–8.2. There is no counterpart in the surviving portion

these systems make up the body's internal bureaucracy, and that the heart (which thinks, wills, and feels) coordinates their activities in the same way as the emperor keeps his civil service working together. Executive virtue is equally crucial in both spheres. Health is defined in terms of an ultimately political ideal.

These examples show that government depends on patterns that also hold for the cosmos and the body, and that the functional systems of the body make up a bureaucracy. To complete the schema, since the state is a little cosmos, the cosmos must be a civil service writ large. This is clear enough from the first of the astronomical treatises in the Standard Histories, which is in fact entitled The Book of Celestial Offices ("T'ien kuan shu 史記天官書"). Each of the constellations it enumerates turns out to be a department staffed by stars.[9]

Comparisons

Let me summarize a few of the conclusions we have drawn from comparing the relations of macrocosms and microcosms in China with those we find in Greece:

1. Greek culture in the period that concerns us encouraged disagreement and disputation in natural philosophy and science as in every other field; in China the emphasis remained on consensus.

2. For Greeks, whatever other purposes it served, oral disagreement was a tool of competition. Without sinecures or even secure employment, philosophers were teachers. They depended on skill in debate for livelihood and fame. They tended to argue face to face and to expect the public to decide, just as it decided in the assembly or at trials. Even when Greeks agreed that something was the case, they seldom agreed on why or how.

In China people who lived by their knowledge, with few exceptions, expected rulers to support them—as "guests" *(ke* 客) in the local courts of the Warring States and as imperial officials in the Han. They presented their ideas much of the time not to colleagues but rather to their patrons, who expected advice but did not have to act on it, or even to reply to it. This relationship hardly made for lively exchanges, and few are recorded. Disagreements with other scholars were, except for a testy few who tended to have unsatisfactory careers, unimportant by comparison. Open attacks were usually written—and one-sided. Patrons faced with political decisions encouraged parleys on concrete questions, but seldom showed patience for anything resembling intellectual debate.

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of *Huang-ti nei ching tai su* 太素.

9. *Shih chi, chüan* 27.

On the whole, the Chinese valued consensus as much as the Greeks valued dispute. In China the relationships of masters and disciples were based on the ritual transmission of written texts.[10] A teacher and his disciples formed an internally cohesive community that avoided attacking other communities. Quarrels were not likely to be productive when teachers aspired above all to official employment for their pupils, and when parents measured success by the same criterion.

3. Like those of China, Greek macrocosms and microcosms reflected political ideals. For the Chinese those ideals remained unifying and centripetal. From the Han on, in a social system that valued civil service above every other career, philosophers who wanted to be politically engaged, or simply respectable, understood the risk of favoring alternatives to the current dispensation of power.

In Greece, with its diverse city-states, constitutions, and political tastes, some people saw the cosmos as a single order, some as an ensemble of quite distinct orders, some as a balance of opposed powers, some as a state of strife. There was no shared ideal to build on.

4. Chinese rulers formed a loyal civil service elite by building on symbols and rituals that literati valued. But this process of alignment bound both. As an element of the new ideology, rulers to varying extents accepted limitations on their power.

The interests of Greek rulers were on the whole irrelevant to the thinkers who developed diverse cosmic metaphors for the state. Philosophers had no voice in the decisions of power-holders. Because intellectuals were not constrained by reasons of state, and because their public roles were played out in disagreement rather than consensus, their stances reflected a great range of contradictory definitions of state as well as of cosmos.

Conclusion

Because of circumstances like the ones I have summed up for this single example, the two cultures differed greatly in what one said and how one said it. The comparison throws into relief the reasons that the discourse of Chinese science was so rarely confrontational. Natural philosophers in the Western Han accepted the state's view of itself because, unlike their Greek counterparts, they formed and maintained it. The same people shaped a view not merely of the political microcosm, but of the cosmos and the body. We intend to explore further the manifold in which the three were united, particularly to understand why it remained desirable to keep them linked as the separate sciences emerged. This sketch of one of our themes suggests that the pieces of the puzzle are beginning to fall into place.

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10. For detailed discussions see Nakayama 1984 and Sivin 1995.